

ROLE OF THE CARDIAC CYCLE IN THE PERCEPTION OF REALITY AND ILLUSION

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References and Acknowledgements



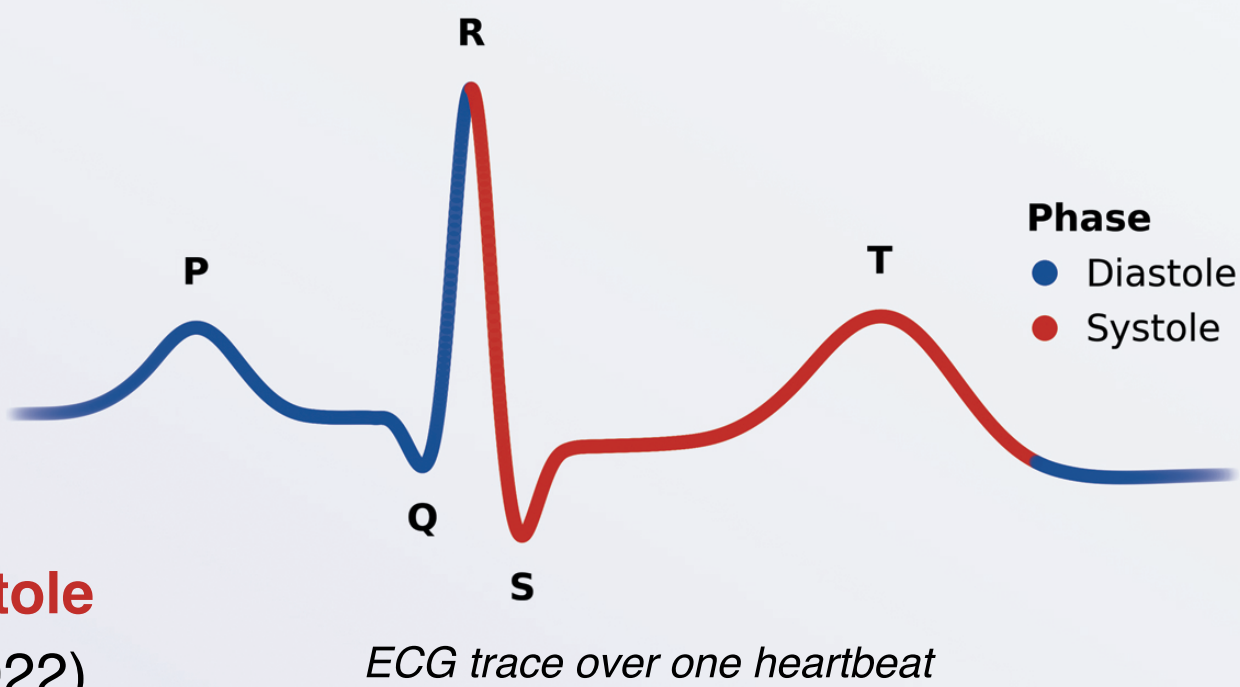
1. BACKGROUND

Over one heartbeat, the heart completes the **cardiac cycle**, comprising **diastole** and **systole**

Signals from the heart seem to play an important role in perception and cognition (Azzalini et al., 2019)

In particular, signals transmitted to the brain during **systole** may reduce the accuracy of perception (Skora et al., 2022)

To understand whether our sense of “reality” may be affected by cardiac signals, we can measure how strongly people perceive **visual illusions** (VIs) to quantify one’s “**VI sensitivity**”. Possessing a higher VI sensitivity means that an individual views a stimulus as having a subjectively stronger illusory effect



2. AIMS AND OBJECTIVES

Aim: To investigate whether the cardiac cycle is related to the perception of VIs

Hypothesis: In the “Illusion Game” task, VI sensitivity will be stronger (more error-prone) when participants view VIs during **systole** (relative to **diastole**)

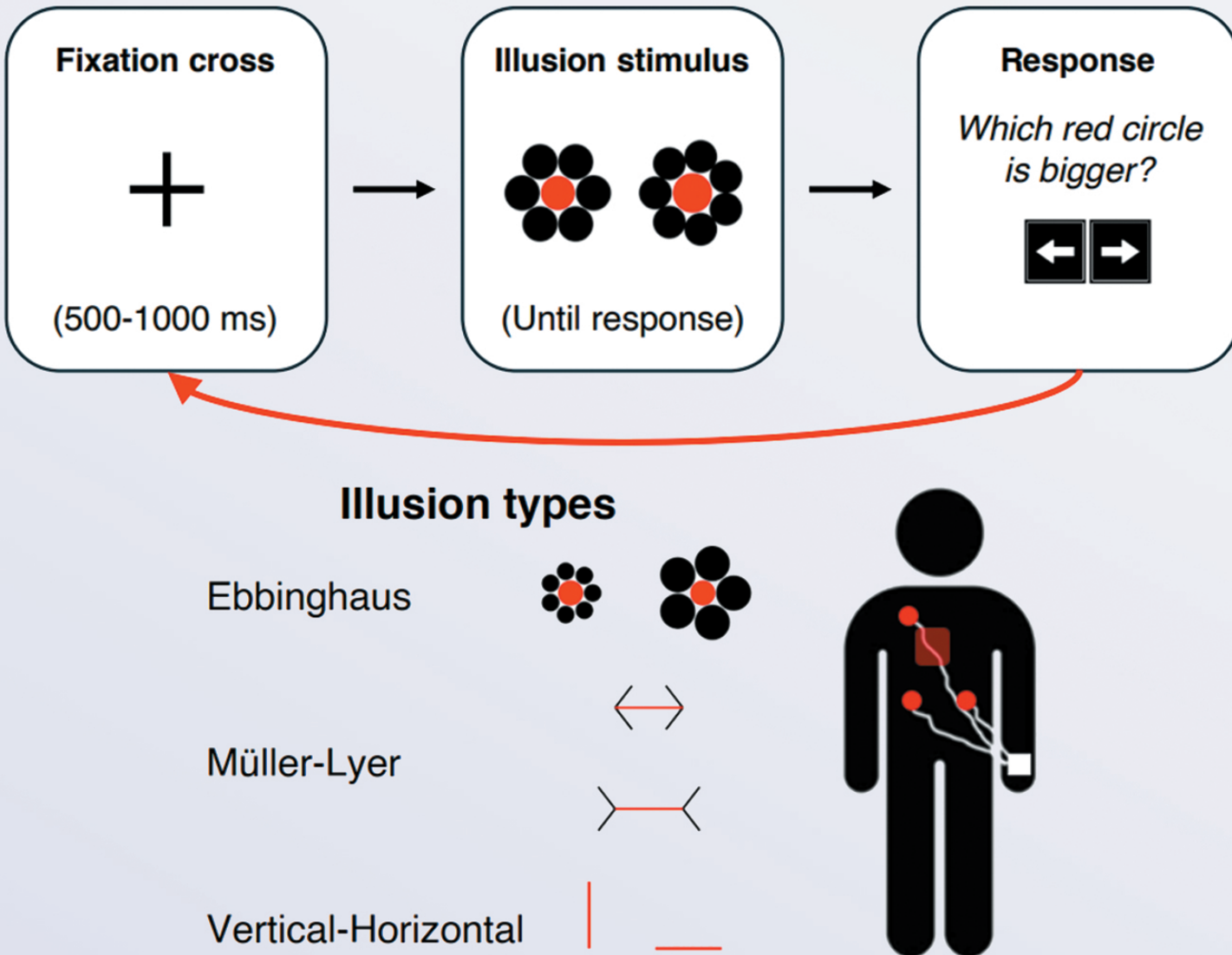
3. PARTICIPANTS

33 participants ($M_{Age} = 25.2$, $SD = 8.7$, range: [18, 66], 57.6% ♀) were recruited at the University of Sussex

4. METHODS

Participants completed the **Illusion Game** (Makowski et al., 2023) whilst their cardiac activity was measured using a 3-lead **electrocardiogram** (ECG)

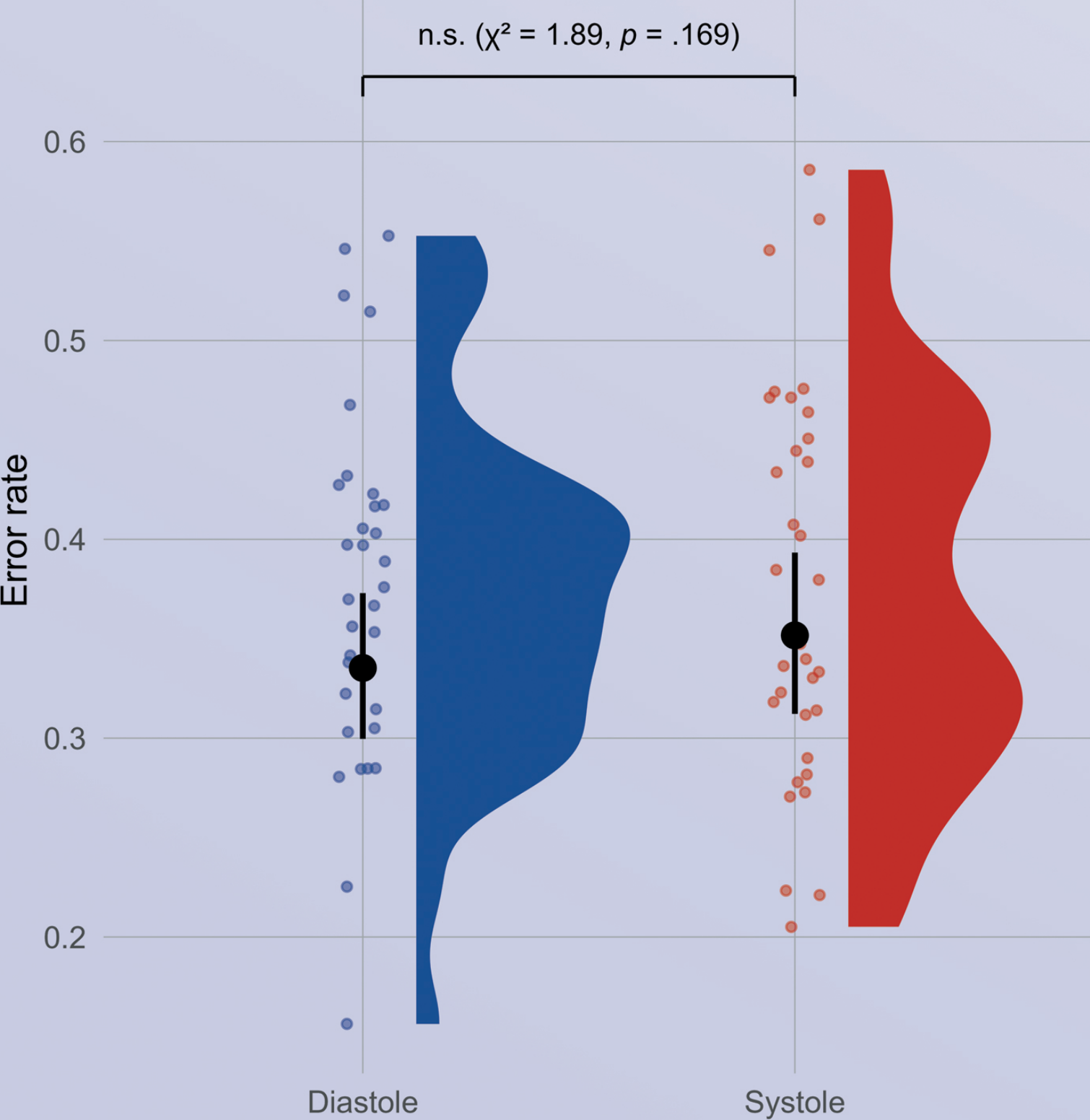
- In this speed-accuracy task, participants made perceptual judgements across three illusion types: **Ebbinghaus**, **Müller-Lyer**, and **Vertical-Horizontal** (e.g., for the Ebbinghaus illusion, participants answered “Which red circle is bigger?” by making correct/incorrect arrow key responses)
- Over 480 trials, **task difficulty** (core illusion elements) and **illusion strength** (illusory context) of the stimuli were manipulated independently and continuously
- Trials were either **congruent** or **incongruent**, i.e., influencing judgements towards or against the correct alternative, respectively. Only **incongruent** trials are included in the analysis
- To quantify VI sensitivity, we can compute a participant’s average **error rate** ($\uparrow$ VI Sens. = $\uparrow$ Errors)



5. RESULTS

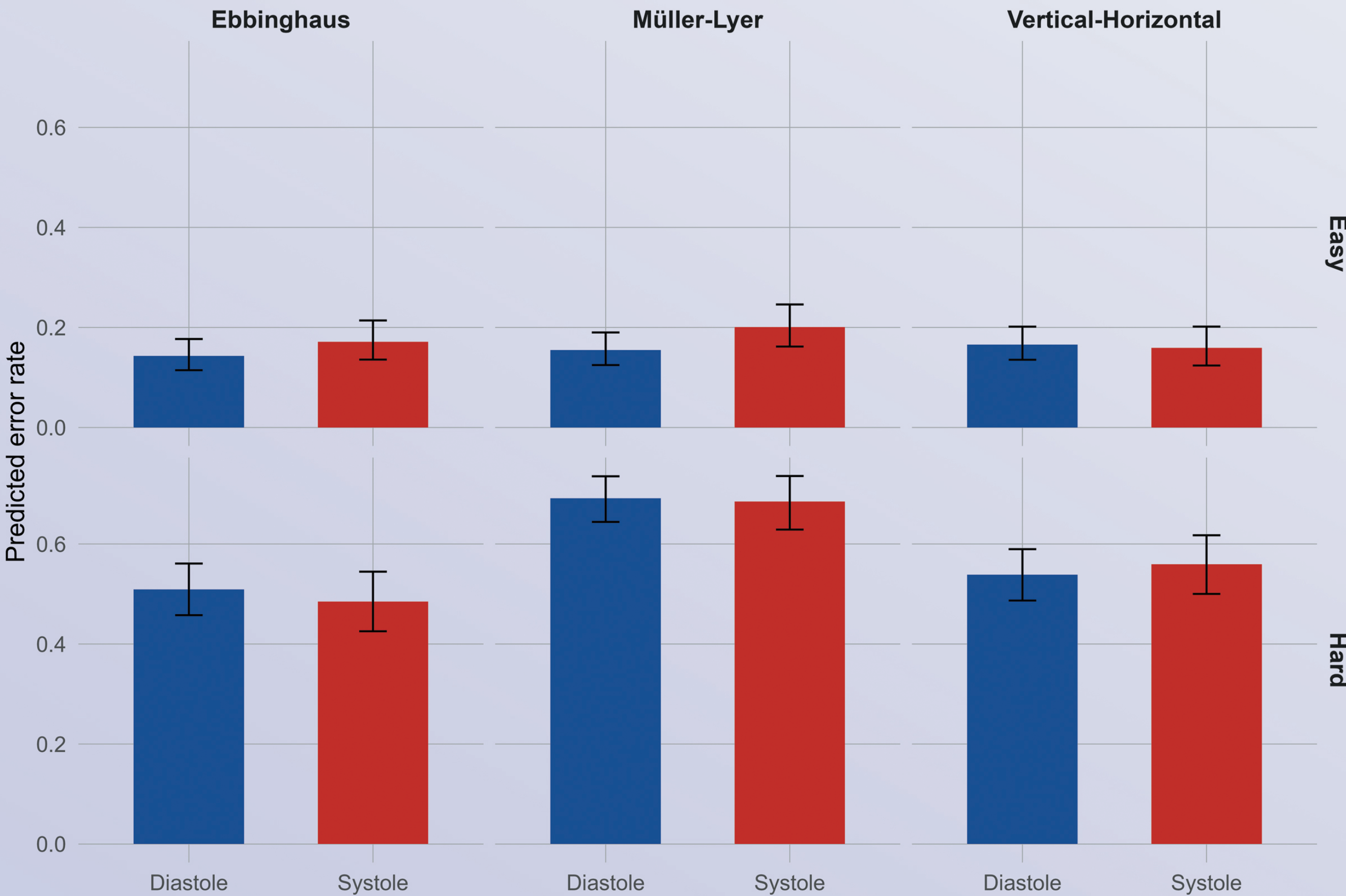
Overall effect of cardiac phase on error rate

Averaged across illusion type and difficulty



Cardiac phase effect by illusion type and difficulty

Cardiac×Illusion: n.s. ($\chi^2 = 3.07$, $p = .216$) | Cardiac×Difficulty: n.s. ($\chi^2 = 2.56$, $p = .109$) | 3-way: n.s. ($\chi^2 = 3.81$, $p = .149$)



Statistical modelling (logistic regression) found **no** evidence that cardiac phase influenced **error rate**, either overall, $\chi^2(1) = 1.89$, $p = .169$, or in interaction with illusion type, $\chi^2(2) = 3.07$, $p = .216$, difficulty, $\chi^2(1) = 2.56$, $p = .109$, or the three-way combination of phase, illusion type, and difficulty, $\chi^2(2) = 3.81$, $p = .149$

6A. DISCUSSION

Overall, the results **fail** to support the hypothesis that VI sensitivity is stronger in **systole** (versus **diastole**)

- At face value, this would suggest that VI sensitivity may **not** be modulated by cardiac phase effects
- However**, it is important to recognise that this sample only consisted of 33 participants, which may constitute insufficient statistical power to detect such a small effect of the cardiac cycle
- Additionally**, cardiac phase was coded as categorical (**diastole/systole**) rather than continuous (**0-100%**), potentially masking variability in cardiac phase dynamics on VI sensitivity

6B. FUTURE DIRECTIONS

We aim to carry out the following:

- Recruit 100 total participants
- Develop more complex computational models to simulate decision processes underlying VIs
- Include measures pertaining to interoception, cognitive control, and cardiac cycle % completed